

Two composable standards. One DSO reference.

The proposed Credentials and DID standards separate credential lifecycle from identity and service discovery.

EARLY DRAFT · CREDENTIALS

Issue and retain

Define the credential contract and lifecycle; registration adds an optional retention view.

EARLY DRAFT · DID

Identify and discover

Resolve a DID Document to find service endpoints. Verification and API trust remain separate.

PROJECT DELIVERABLE

DSO reference

Build the DSO Credential Registry and DSO DID Registry together. Standards adopters need not run either.

PR #204 proposed a registry with ANS / Scan discovery. CNS / ANS offers an existing name-to-Party lookup anchor, not the scope of these drafts. No production migration is defined.

One credential. Registration adds a separate lifecycle.

`Credential` and `RegisteredCredential` are views of the same contract, not competing identities.

INTRINSIC CONTRACT

Credential

Holds the intrinsic credential data and validity. It can exist without a registry.

`validUntil` governs credential validity.

OPTIONAL REGISTRY VIEW

RegisteredCredential

Exposes registration for that credential, when registration is chosen.

`expiresAt` governs registry retention.

One canonical contract; the registered view adds registry metadata. DSO paid retention would extend `expiresAt`, never `validUntil`. Payment and authorization for that extension remain open.

The W3C issuer may be external. The anchor is on Canton.

ISSUER

W3C identifier

The issuer identifier need not be a Canton Party. An external identifier does not become a Canton signatory.

ANCHORING

Canton parties

Anchorers are mandatory Canton parties responsible for anchoring the credential.

HOLDING

Optional holders

Holders are optional Canton parties. The ledger contract remains Canton-specific.

Signatories: anchorers plus holders, deduplicated. Anchoring records a claim; it does not verify the claim. Encoding issuer or holder as a mandatory Canton Party in downstream templates would force migration or a false Party identity.

Who holds it? Who is it about?

Alice holds an ACME licence. Alice is the holder; ACME is the explicit credential subject.

OLD KEY CONVENTION

Subject hidden in a key

`licenseStatus!ACME` identifies ACME by convention; bare `licenseStatus` implies Alice. This conflates claims and subject selection.

DRAFT MODEL

Subject is explicit

Set the credential subject to ACME and use `licenseStatus` as the claim key. Holder Alice remains a separate role.

A compatibility projection for old key prefixes is a **proposal**, not implemented behavior. Do not infer the subject from a claim key in the new model.

Resolve a DID. Find its services.

A DID Document binds identity to discovery metadata. Explicit service entries point clients to versioned API URLs.

01

DID identifier

Identify the subject; controller and binding rules depend on the DID method.

02

DID Document

Publish identity state and typed service entries under method rules.

03

Resolver

Resolve and verify DID state before using discovered endpoints.

04

Service URLs

Discover one or more Credentials API endpoints; access policy is separate.

CNS / ANS already supports name-to-Party lookup; Scan projects DSO stakeholder data. As a proposed example, a CNS name could also be represented in a credential, but that is not required by either standard.

Issue and renew. Register and retain separately.

Two clocks govern different things; neither requires a transaction merely because time passes.

INTRINSIC LIFECYCLE

Issue and renew

Factory issuance creates a credential. Renewal changes `validUntil` only: credential validity, not registry retention.

REGISTRY LIFECYCLE

Register, extend, archive

Registration is optional. Extension changes `expiresAt` only: registry retention, not credential validity. Archiving is separate.

Selected Daml lifecycle POC: compare these transitions. Offer / Accept, AddHolder, suspension, resumption and revocation remain open, **not delivered**. External off-ledger W3C proof verification is future work, not implemented in Daml.

Two resources. Active and archived reads.

The selected PQS POC tests API reads, not complete ledger choice history.

HTTP RESOURCES

Intrinsic credential

Expose independently of registration; no registry record required.

HTTP RESOURCES

Registered credential

Separate resource; credential data and registration metadata stay distinct.

PQS READS

Active records

Fixed-offset, keyset-paginated reads.

PQS READS

Archived records

Fixed-offset, keyset-paginated reads.

The PQS POC does **not** promise complete choice-event replay. Reads and lifecycle are separate concerns.

TAKEAWAY

Separate the contract, the registry and discovery.

Build the DSO Credential Registry and DID Registry as a reference implementation of two composable early drafts.

CREDENTIAL

Intrinsic truth

Explicit subject and roles; validity belongs to the credential.

REGISTRATION

Optional retention

Registry metadata and retention have their own lifecycle.

DID + API

Discover, then verify

Discovery points to an API; trust and read guarantees need separate decisions.

Selected POCs test lifecycle, PQS reads and multi-endpoint DID resolution. None alone delivers combined DSO BFT or payment.